

Lists, dictionaries, sets, and tuples

Module 2 - Lesson 2

Overview

Collections hold many values. Choosing the right container is a design decision: lists are ordered, dicts map keys to values, sets store unique items, and tuples are fixed.

Key Concepts

- Lists are ordered and mutable: append, extend, insert, remove, sort, and slicing.
- Dicts map hashable keys to values; `d[k]` or `d.get(k, default)` read them safely.
- Sets store unique, unordered items; `&` | - perform set algebra.
- Tuples are immutable sequences often used for fixed records and function returns.
- Choose the container by the operation: lookup -> dict, uniqueness -> set, order -> list.
- Comprehensions build new collections concisely from existing iterables.

Example

```
grades = [88, 92, 79]
summary = {
    "max": max(grades),
    "passed": all(g >= 50 for g in grades),
}
unique_tags = {"python", "web", "python", "api"}
print(unique_tags) # {'web', 'python', 'api'}
```

Summary

Lists, dicts, sets, and tuples each model a different shape of data. Matching the container to the problem makes code shorter, safer, and faster.

Practice Checklist

- Remove duplicates from a list using a set.
- Build a dict that counts the frequency of each letter in a word.
- Use a tuple to return multiple values from a function.